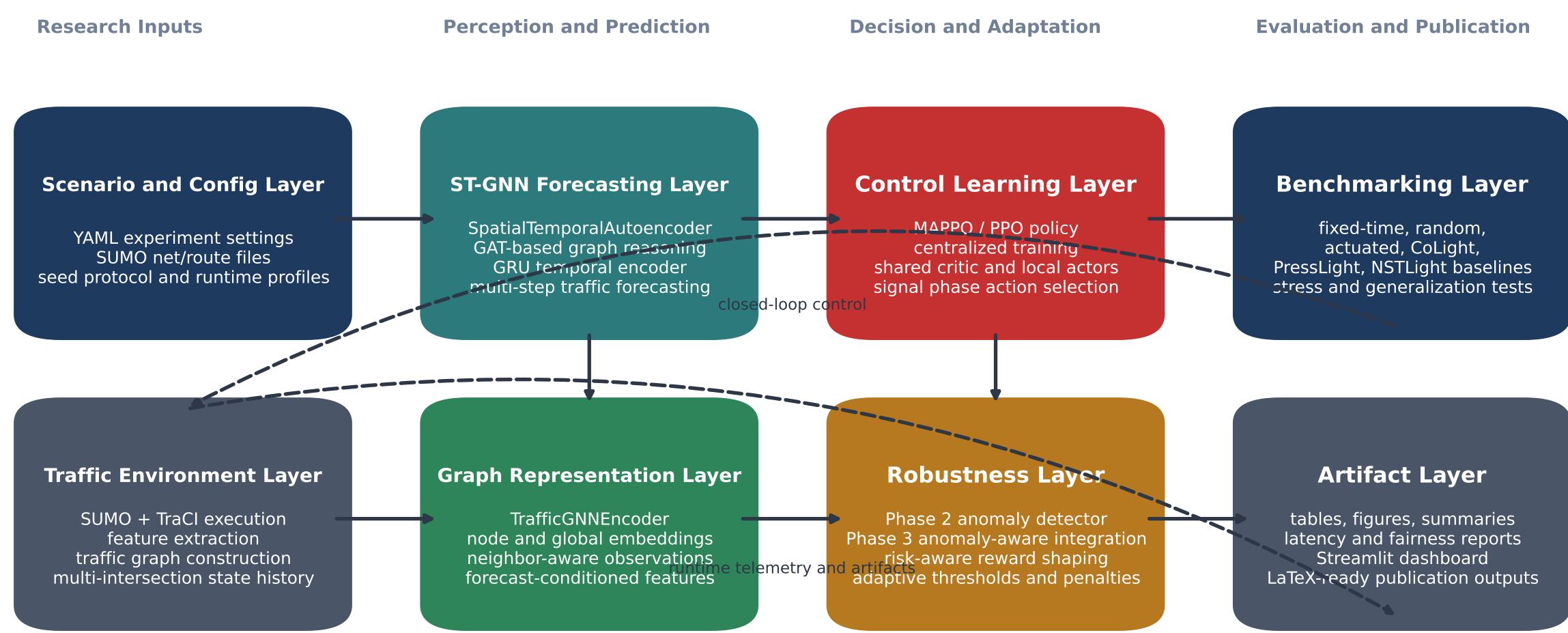


Whole-Framework Architecture for Robust Multi-Agent Traffic Control



Closed-loop operation: scenario configuration initializes SUMO, the ST-GNN produces forecast-aware embeddings, MAPPO selects decentralized signal actions, and the evaluation layer converts runtime outcomes into reproducible publication artifacts.